

Wildcard Masking

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Wildcard Mask Overview

- Tells a router exactly which bits of an IP address must match, and which ones it can ignore
- Used in ACLs to target a single address, a range of addresses, or every address
- Written like an IP address — four numbers separated by dots
- 0 = “this bit must match”, 1 = “ignore this bit” — the opposite of a subnet mask

Example:

The wildcard mask 0.0.0.0 means every bit must match — it points to one exact address only.

Last Octet of Wildcard 0.0.0.15



= 15 (decimal)



0 = must match



1 = can vary

These last 4 bits can be anything, so this wildcard matches 16 addresses in a row (.0 through .15).

Wildcard Mask Types

Match One Host

0.0.0.0

Every bit must match exactly

Example:

host 192.168.1.5

Match a Range

0.0.0.15

Last few bits can vary

Example:

192.168.1.0 0.0.0.15

Match Any Address

255.255.255.255

Every bit can vary

Example:

any

Wildcard Mask Calculation

- A wildcard mask is calculated by subtracting the subnet mask from 255.255.255.255
- This flips which bits matter — the network bits become “ignore” bits, and vice versa
- Used when writing an ACL rule that needs to match a whole subnet, not just one address

Example:

For subnet mask 255.255.255.0 (a /24 network), the wildcard mask works out to 0.0.0.255.

255.255.255.255

– 255.255.255. 0

0. 0. 0.255

wildcard mask for /24

Wildcard Mask Keywords

- Cisco IOS gives two shortcuts, so you don't have to type long wildcard masks
- `host` — shortcut for wildcard `0.0.0.0` → matches one exact address
- `any` — shortcut for wildcard `255.255.255.255` → matches every address

Example:

`access-list 10 permit host 192.168.1.5`

same as: `access-list 10 permit 192.168.1.5 0.0.0.0`

`access-list 10 permit any`

same as: `access-list 10 permit 0.0.0.0 255.255.255.255`

Summary

- A wildcard mask tells a router which bits must match, and which bits it can ignore
- 0 = must match, 1 = ignore — the reverse of a subnet mask
- Calculated as 255.255.255.255 minus the subnet mask
- host and any are shortcuts for the two most common wildcard masks
- Wildcard masks let one ACL rule target a single address, a range, or everyone

References

Cisco Networking Academy

<https://www.netacad.com>

Cisco

<https://www.cisco.com/c/en/us/support/docs/ip/access-lists/26448-ACLsamples.html>

NIST

<https://www.nist.gov/cyberframework>